

Logistic regression interpretation

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Logistic regression model

Y_i = dengue status (0 = negative, 1 = positive)

Age_i = age (in years)

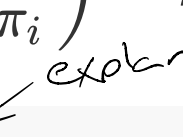
Random component: $Y_i \sim \text{Bernoulli}(\pi_i)$

Systematic component: $\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 Age_i$

Fitting the logistic regression model

$$Y_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{Age}_i$$

response  *explanatory* 

```
m1 <- glm(Dengue ~ Age, data = dengue,  
          family = binomial)  
summary(m1)
```

tell us distribution of response
for logistic, use binomial family

Fitting the logistic regression model

$$Y_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{Age}_i$$

```
m1 <- glm(Dengue ~ Age, data = dengue,
           family = binomial)
summary(m1)
```

...

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
## (Intercept)	-2.454345	0.075068	-32.70	<2e-16 ***
## Age	0.217312	0.008826	24.62	<2e-16 ***

...

fitted model:

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{Age}_i$$

z-statistic, not
t-statistic

Last time: Activity

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 \overset{70}{+ 0.22} Age_i$$

Does the probability of dengue increase or decrease with age?

increases!

Activity

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

What are the predicted odds of dengue for a 5 year old patient?

$$\begin{aligned} -2.45 + 0.22(5) &= -1.35 \\ &\quad \uparrow \text{log odds} \\ \text{odds} &= e^{-1.35} \approx 0.26 \\ \nearrow \frac{\hat{\pi}_i}{1 - \hat{\pi}_i} \end{aligned}$$

Activity

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

For a 5 year old patient: is the predicted *probability* of dengue > 0.5 , < 0.5 , or $= 0.5$?

if $\hat{\pi}_i < 0.5$, then

$$\Rightarrow \frac{\hat{\pi}_i}{1 - \hat{\pi}_i} < 1$$

if $\hat{\pi}_i = 0.5$, then

$$\frac{\hat{\pi}_i}{1 - \hat{\pi}_i} = 1$$

if $\hat{\pi}_i > 0.5$, then

$$\frac{\hat{\pi}_i}{1 - \hat{\pi}_i} > 1$$

here odds $\approx 0.26 < 1$, so $\hat{\pi}_i < 0.5$

Calculating probability

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

$$\begin{aligned} \text{odds} &= \frac{\text{prob}}{1 - \text{prob}} \\ \text{prob} &= \frac{\text{odds}}{1 + \text{odds}} = \frac{e^{B_0 + B_1 X_i}}{1 + e^{B_0 + B_1 X_i}} \end{aligned}$$

Predicted *probability* of dengue for a 5 year old patient?

$$\text{odds} = \frac{\hat{\pi}_i}{1 - \hat{\pi}_i} = 0.26$$

$$\hat{\pi}_i = 0.26 (1 - \hat{\pi}_i)$$

$$\Rightarrow \hat{\pi}_i = 0.26 - 0.26 \hat{\pi}_i$$

$$\Rightarrow \hat{\pi}_i + 0.26 \hat{\pi}_i = 0.26$$

$$\Rightarrow \hat{\pi}_i (1 + 0.26) = 0.26$$

$$\Rightarrow \hat{\pi}_i = \frac{0.26}{1 + 0.26} \approx 0.21$$

Activity

Work on the activity (handout) with a neighbor, then we will discuss as a class.

Activity

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

What is the predicted probability of dengue for a 10 year old patient?

$$\exp\{-2.45 + 0.22(10)\} \approx 0.78$$

$$\hat{\pi}_i = \frac{\text{odds}}{1 + \text{odds}} = \frac{0.78}{1 + 0.78} \approx 0.44$$

Activity

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

Suppose we want to identify patients for whom the predicted probability of dengue is at least 0.5. What age range should we focus on?

$$\hat{\pi}_i = \frac{e^{-2.45 + 0.22 \text{ Age}_i}}{1 + e^{-2.45 + 0.22 \text{ Age}_i}} \stackrel{?}{>} 0.5$$

$$\hat{\pi}_i \geq 0.5 \Leftrightarrow \text{odds} > 1 \Leftrightarrow \log \text{odds} > 0$$

$$-2.45 + 0.22 \text{ Age} > 0$$

$$\Rightarrow \text{Age} > 11.14$$

$$\boxed{11.14 \leq \text{Age} \leq 15}$$

In the study, only observed patients up to 15 yrs old
(don't want to extrapolate)

Activity

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

Compare the odds of dengue for a 12 year old patient to the odds of dengue for an 11 year old patient. What do you notice?

$$\text{Odds ratio (OR)} = \frac{\text{odds when age} = 12}{\text{odds when age} = 11} \approx 1.25$$

$$= \frac{\text{odds when age} = 11}{\text{odds when age} = 10}$$

In general:

$$\frac{\text{odds when age} = x+1}{\text{odds when age} = x} = \frac{e^{-2.45 + 0.22(x+1)}}{e^{-2.45 + 0.22x}} = e^{0.22} \approx 1.25$$

Interpretation

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

Slope:

$$e^{0.22} = 1.25 \Rightarrow \text{multiply odds by 1.25 for each extra year in Age}$$

- + A one-year increase in age is associated with an increase of 25% in the *odds* of dengue.

Intercept:

Ex: $\hat{\beta}_1 = -0.15$

$$\Rightarrow e^{\hat{\beta}_1} = e^{-0.15} = 0.86$$

$\Rightarrow$ 14% decrease in odds

Interpretation

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

Slope:

- + A one-year increase in age is associated with an increase of 25% in the *odds* of dengue.

Intercept:

- + when age is 0, the estimated odds of dengue is $e^{-2.45} = 0.086$

Can we interpret coefficients in terms of probability?

Age 10

Age = 11

Age = 12

log odds	-0.25	$\xrightarrow{+0.22}$	-0.03	$\xrightarrow{+0.22}$	0.19
odds	0.78	$\xrightarrow{\times 1.25}$	0.97	$\xrightarrow{\times 1.25}$	1.21
$\hat{\pi}_i$	0.44	$\xrightarrow{+0.05}$ $\times 1.11$	0.49	$\xrightarrow{+0.06}$ $\times 1.12$	0.55


different!

no constant change in probability for a one-year increase in Age

can't interpret coefficients in terms of probability changes

Adding more variables

Now let's add WBC as a variable to the model:

```
m2 <- glm(Dengue ~ Age + WBC, data = dengue,  
           family = binomial)  
summary(m2)
```

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = 0.34 + 0.15 \text{ Age}_i - 0.31 \text{ WBC}_i$$

How should I interpret each coefficient in the fitted model?

0.15: $e^{0.15} = 1.16$

A one-year increase in Age is associated w/ an increase in odds of dengue by 16%, holding WBC fixed

Adding more variables

What about a *categorical* predictor?

```
m3 <- glm(Dengue ~ Age + Abdominal, data = dengue,  
          family = binomial)  
summary(m3)
```

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -0.25 + 0.22 \text{ Age}_i + 0.21 \text{ Abdominal}_i$$

How should I interpret the coefficient on abdominal pain?

For next week

Reading: in the textbook, read

- + sections 6.7.1 - 6.7.3
- + sections 2.2 - 2.4